The quadratic formula

WORKSHEET

Interactive version

Name _____ Class _____ Date _____

WORKED EXAMPLES

Three worked examples

Example 2 · The second garden: x² + 3x − 30 = 0

COEFFICIENTS
$a = 1,\ b = 3,\ c = -30.$

DISCRIMINANT
$D = 3^2 - 4 \cdot 1 \cdot (-30) = 9 + 120 = 129 > 0.$

FORMULA
$x_{1,2} = \dfrac{-3 \pm \sqrt{129}}{2}$, so $x_1 \approx 4.18$ and $x_2 \approx -7.18$.

ANSWER · YOUR TURN

_____

Example 3 · One root: 4x² − 4x + 1 = 0

COEFFICIENTS
$a = 4,\ b = -4,\ c = 1.$

DISCRIMINANT
$D = (-4)^2 - 4 \cdot 4 \cdot 1 = 16 - 16 = 0$, so there is one root.

ROOT · YOUR TURN

_____

CHECK · YOUR TURN

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CHECK

Check yourself

Level 1 · threshold

1.

How many real roots does $3x^2 - 5x + 4 = 0$ have?

(A) one
(B) none
(C) can't tell without solving
(D) two

Level 1 · threshold

2.

Solve $x^2 - 5x - 14 = 0$. Give both roots, separated by a semicolon.

Answer:

Level 2 · satisfactory

3.

Put the steps for solving $2x^2 + x - 6 = 0$ in the right order.

- [ ] Read off $a = 2$, $b = 1$, $c = -6$.
- [ ] Compute $D = 1^2 - 4 \cdot 2 \cdot (-6) = 49$.
- [ ] $x_{1,2} = \frac{-1 \pm 7}{4}$.
- [ ] $D > 0$, so there are two roots.
- [ ] $x_1 = \frac{3}{2}$, $x_2 = -2$.

Level 2 · satisfactory

4.

For which value of c does $x^2 + 6x + c = 0$ have exactly one root?

Answer:

Level 3 · basic

5.

Which equations have two different real roots? Choose all that apply.

(A) $x^2 - 4x + 4 = 0$ (B) $x^2 + 4 = 0$ (C) $-x^2 + 3x + 1 = 0$ (D) $x^2 - 4 = 0$

EXAM TASK

Two consecutive integers

Level 3 · basic PUPP style · 4 points

The sum of the squares of two consecutive positive integers is 181. Find the integers.